Method	Memory	Budget	Mean scan time	MAPE % [95% CI]	Scans < 5%
RL	**TI histogram**	**560 s**	**541 s**	**2.93 [2.58, 3.30]**	**95.8%**
RL	none	560 s	548 s	4.16 [3.41, 4.99]	66.7%
RL	fitter uncertainty (σ)	560 s	544 s	5.25 [4.32, 6.22]	58.3%
RL	LSTM[†]	560 s	525 s	7.31 [6.10, 8.65]	25.0%
Fixed log TI grid	–	560 s	504 s	6.04 [5.36, 6.70]	25.0%
Cramér-Rao optimal	–	560 s	558 s	7.74 [7.08, 8.42]	8.3%

[†] LSTM: selected checkpoint came from the cached rung, before any full-Bloch training; the run had 7.3× fewer full-Bloch steps than the TI-histogram run (37,862 vs 274,537).